PART 2

Inspection SOP

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1. Purpose

Defines scheduled inspection of PLC-FIL-02, its signal modules, communication modules, wiring and supporting equipment, to find faults before they stop production.

2. Frequency

Level	Frequency	Performed by	Typical duration
A - Visual walk-around	Weekly	Maintenance technician	10 min
B - Functional check	Quarterly	Maintenance technician	45 min
C - Full inspection and backup	Annually (planned shutdown)	Technician + controls engineer	3 h

3. Tools and materials

- Laptop with STEP 7 (TIA Portal) and Ethernet patch cable
- Multimeter (calibrated), insulated screwdrivers, torque screwdriver
- Infrared thermometer or thermal camera
- Lint-free cloth, contact cleaner (non-residue), ESD wrist strap
- Spare SIMATIC memory card and battery board (if fitted)

4. Level A - Visual walk-around (weekly)

#	Check	Acceptance criterion	OK / Fail
A1	CPU status LEDs	RUN/STOP green, ERROR off, MAINT off	
A2	Signal module and communication module LEDs	No red or abnormal flashing states	
A3	Enclosure condition	Door sealed, no dust build-up, no moisture, no unusual odour	
A4	Cabinet temperature / ventilation	Filters clean, fans running, within site limit	
A5	Cables and connectors	No damage, strain, or loose connectors	
A6	HMI communication	HMI shows live data, no comms alarm	

5. Level B - Functional check (quarterly)

#	Check	Acceptance criterion	OK / Fail
B1	Diagnostic buffer review (STEP 7 or Web server)	No unexplained errors since last inspection; log reviewed and entries explained	
B2	Cycle time and memory usage	Max cycle time well below configured watchdog; load and work memory within agreed headroom	
B3	Real-time clock	Offset from plant time source within 5 s; correct time zone	
B4	24 VDC supply voltage at CPU	Within 20.4 to 28.8 V DC (CPU 1214C DC specification range)	
B5	Terminal tightness (de-energised, under LOTO)	No loose terminals; torque per site standard	
B6	Sample I/O tests	Test 3 critical inputs and 3 outputs via watch table without forcing	

#	Check	Acceptance criterion	OK / Fail
B7	PROFINET link and IP check	LINK LED on, ping test OK, no duplicate IP or device-name alarms	
B8	Memory card	Fitted, seated, no card error LED pattern	

6. Level C - Full inspection and backup (annual)

- Perform all Level A and B checks.
- Take a full STEP 7 project archive and a CPU upload. Verify offline versus online compare shows no differences, and store both in the project archive.
- Check firmware versions of CPU and modules against the approved baseline and record in Part 4.
- Verify the real-time clock backup (battery board, where fitted) and replace per its schedule.
- Review access protection settings and confirm the password vault entry is current.
- Check thermocouple/RTD/analog module scaling against a calibrated source for critical loops.
- Review spare-parts stock: CPU, one of each signal module type, memory card, fuses, 24 V supply.

7. Findings classification

Class	Definition	Required action
Critical	Fault that can stop or endanger the line now (CPU ERROR solid red, supply failure)	Escalate immediately per POL-PLC-003, Level 3
Major	Degraded function or repeat fault (intermittent comms, repeated diagnostic errors)	Work order within 24 h; notify controls engineer
Minor	Housekeeping or cosmetic (dust, label wear)	Fix at next routine visit

Sign-off: Inspector _____ Date _____ | Reviewer _____ Date _____